

Homework IoT - Exercise 3

Mattia Brianti 10773859

Alex Hathaway 10829946

A RFID system based on Dynamic Frame ALOHA is composed of $N=4$ tags.

1 Question 1

Find the overall collision resolution efficiency η in the different cases in which the initial frame size is set to $r_1=1,2,3,4,5,6$.

- Assume that after the first frame, the frame size is correctly set to the current backlog size
- Assume as given the duration of the arbitration period with $N=2,3$ tags when $r=N$ ($L_2 = 4$, $L_3 = 51/8$)

The collision resolution efficiency η can be calculated as the number of tags divided by the expected total number of time slots it takes to read them. Since the number of tags is fixed and is equal to 4 the goal is to find $L_4(r_1)$, which is the expected number of total slots required to solve all 4 tags, given a specific starting frame size of r_1 :

$$\eta = \frac{N}{L_N(r_1)} = \frac{4}{L_4(r_1)} \quad (1)$$

Now we calculate the expected time. In Dynamic Frame ALOHA, the frame size adapts to the backlog, which means the expected time to solve everything is the time spent in the current frame plus the expected time to resolve the collided and failed tags. The general formula for the expected total slots starting with an initial frame r_1 is:

$$L_4(r_1) = r_1 + \sum_{s=0}^3 P(S=s) \cdot L_{4-s}^* \quad (2)$$

Where $P(S=s)$ is the combinatorial probability that exactly s tags successfully transmit and L_{4-s}^* is the baseline expected time for a frame perfectly matched to 4 tags ($r_1 = 4$).

To find the efficiency values we must start by calculating the probabilities. Because every tag makes an independent choice, the total number of ways N tags can arrange themselves into r slots is r^N . In our case the total possible arrangements are $4^4 = 256$.

- **0 Successes ($S=0$):** All tags collide.
 - 4 tags in 1 slot: 4 ways.
 - 2 tags in one slot, 2 in another: $\binom{4}{2} \times \frac{1}{2} \binom{4}{2} \times 2 = 36$ ways.
 - Total = 40 ways. $\Rightarrow P(S=0) = \frac{40}{256} = \frac{5}{32}$
- **1 Success ($S=1$):** 3 tags in one slot, 1 isolated tag.
 - $4 \times 3 = 12$ slot choices $\times 4$ tag choices = 48 ways.
 - Total = 48 ways. $\Rightarrow P(S=1) = \frac{48}{256} = \frac{3}{16}$
- **2 Successes ($S=2$):** 2 tags in one slot, 2 isolated tags.
 - 4 collision slots $\times \binom{4}{2}$ colliding tags $\times 3$ remaining slots $\times 2$ ways to place isolated tags = 144 ways.
 - Total = 144 ways. $\Rightarrow P(S=2) = \frac{144}{256} = \frac{9}{16}$

- **4 Successes ($S = 4$):** 1 tag per slot.
 - $4! = 24$ ways.
 - Total = 24 ways. $\Rightarrow P(S = 4) = \frac{24}{256} = \frac{3}{32}$

Because the system adapts, 0 successes requires re-arbitrating all 4 tags, creating a recursive equation:

$$\begin{aligned}
 L_4^* &= 4 + P(S = 0)L_4^* + P(S = 1)L_3^* + P(S = 2)L_2^* + P(S = 4)L_0^* \\
 L_4^* &= 4 + \frac{5}{32}L_4^* + \frac{3}{16}(6.375) + \frac{9}{16}(4) \\
 L_4^* \left(1 - \frac{5}{32}\right) &= 4 + 1.1953 + 2.25 \\
 L_4^* &= \frac{7.4453}{0.84375} \approx 8.824 \text{ slots}
 \end{aligned}$$

Now that we have isolated the baseline expected time L_4^* for a backlog of 4 tags, we can calculate the expected time $L_4(r_1)$ for any other initial frame size.

For each of the following initial frame sizes, we first determine the probability distribution of successes $P(S = s)$ by evaluating the favorable combinatoric distributions of 4 tags over r_1 slots, divided by the total possible outcomes (r_1^4). We then plug these probabilities into the non-recursive expected time formula:

$$L_4(r_1) = r_1 + P(S = 0)L_4^* + P(S = 1)L_3^* + P(S = 2)L_2^* + P(S = 4)L_0^*$$

Finally, the resulting $L_4(r_1)$ is substituted into Equation 1 to find the efficiency $\eta(r_1)$.

For $r_1 = 1$

4 tags into 1 slot. Total outcomes = $1^4 = 1$.

- $P(S = 0) = 1$

$$L_4(1) = 1 + (1 \times 8.824) = 9.824 \text{ slots} \quad \Rightarrow \quad \eta(1) = \frac{4}{9.824} \approx \mathbf{0.4072} \quad (3)$$

For $r_1 = 2$

4 tags into 2 slots. Total outcomes = $2^4 = 16$.

- $S = 0$ (4 in one, or 2 and 2): $2 + 6 = 8$ ways. $P(S = 0) = \frac{8}{16} = 0.5$
- $S = 1$ (3 in one, 1 isolated): 8 ways. $P(S = 1) = \frac{8}{16} = 0.5$

$$L_4(2) = 2 + 0.5(8.824) + 0.5(6.375) = 9.5995 \text{ slots} \quad (4)$$

$$\eta(2) = \frac{4}{9.5995} \approx \mathbf{0.4167} \quad (5)$$

For $r_1 = 3$

4 tags into 3 slots. Total outcomes = $3^4 = 81$.

- $P(S = 0) = \frac{21}{81} = \frac{7}{27}$
- $P(S = 1) = \frac{24}{81} = \frac{8}{27}$
- $P(S = 2) = \frac{36}{81} = \frac{12}{27}$

$$L_4(3) = 3 + \left(\frac{7}{27} \times 8.824\right) + \left(\frac{8}{27} \times 6.375\right) + \left(\frac{12}{27} \times 4\right) \approx 8.954 \text{ slots} \quad (6)$$

$$\eta(3) = \frac{4}{8.954} \approx \mathbf{0.4467} \quad (7)$$

For $r_1 = 4$

Already calculated as our baseline L_4^* .

$$L_4(4) = 8.824 \text{ slots} \Rightarrow \eta(4) = \frac{4}{8.824} \approx \mathbf{0.4533} \quad (8)$$

For $r_1 = 5$

4 tags into 5 slots. Total outcomes = $5^4 = 625$.

- $P(S = 0) = \frac{65}{625} = 0.104$
- $P(S = 1) = \frac{80}{625} = 0.128$
- $P(S = 2) = \frac{360}{625} = 0.576$
- $P(S = 4) = \frac{120}{625} = 0.192$

$$L_4(5) = 5 + (0.104 \times 8.824) + (0.128 \times 6.375) + (0.576 \times 4) + 0 \approx 9.0377 \text{ slots} \quad (9)$$

$$\eta(5) = \frac{4}{9.0377} \approx \mathbf{0.4426} \quad (10)$$

For $r_1 = 6$

4 tags into 6 slots. Total outcomes = $6^4 = 1296$.

- $P(S = 0) = \frac{96}{1296} = \frac{2}{27}$
- $P(S = 1) = \frac{120}{1296} = \frac{5}{54}$
- $P(S = 2) = \frac{720}{1296} = \frac{5}{9}$
- $P(S = 4) = \frac{360}{1296} = \frac{5}{18}$

$$L_4(6) = 6 + \left(\frac{2}{27} \times 8.824\right) + \left(\frac{5}{54} \times 6.375\right) + \left(\frac{5}{9} \times 4\right) + 0 \approx 9.466 \text{ slots} \quad (11)$$

$$\eta(6) = \frac{4}{9.466} \approx \mathbf{0.4226} \quad (12)$$

2 Question 2

After computing the values of the efficiency with the different frame sizes, produce a plot with values of η over r_1 .

To do so we wrote a small python script and hardcoded the efficiency values previously found.

```

1 import matplotlib.pyplot as plt
2 r_values = [1, 2, 3, 4, 5, 6]
3 efficiencies = [0.4072, 0.4167, 0.4467, 0.4533, 0.4426, 0.4226]
4 plt.figure(figsize=(10, 6))
5 plt.plot(r_values, efficiencies, marker='o', linestyle='--', color='#1f77b4',
6         label='N=4')
7 plt.title("Efficiency  $\eta$  over values of initial frame size  $r_1$ ", fontsize=14)
8 plt.xlabel("Initial frame size  $r_1$ ", fontsize=12)
9 plt.ylabel("Efficiency  $\eta$ ", fontsize=12)
10 plt.xticks(r_values)
11 plt.grid(True, linestyle='--', alpha=0.7)
12 plt.legend()
13 plt.show()

```

The above script generated the following plot:

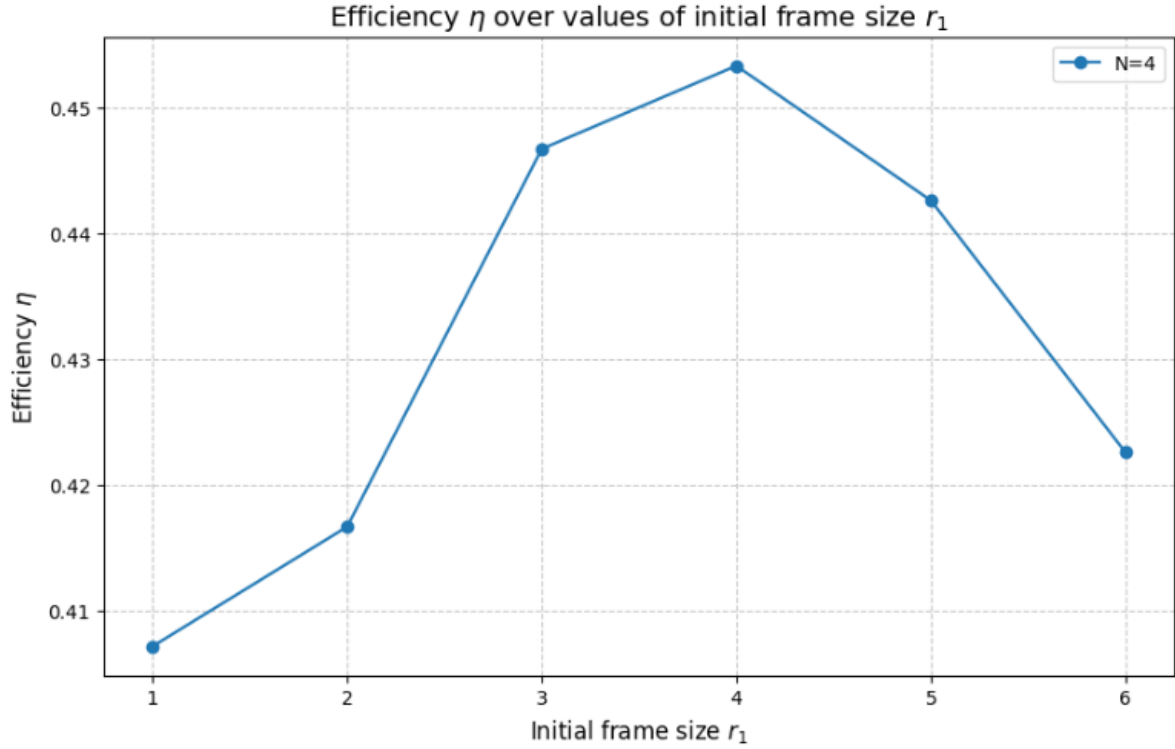


Figure 1: Efficiency over the initial frame size values

3 Question 3

For what values of r_1 we have the maximum value of η ?

The calculations show that the maximum efficiency ($\eta \approx 0.4533$) is achieved when $r_1 = 4$, which corresponds exactly to the number of tags N . This result highlights the fundamental principle of Frame ALOHA systems: efficiency is maximized when $r = N$ because it provides the optimal balance between two types of wasted time: collision and empty slots.

- If $r < N$ (too few slots), the tags are forced into a tight space, making the probability of collisions excessively high.
- If $r > N$ (too many slots), the probability of tags colliding drops, but the system wastes time waiting out empty slots where no tag is transmitting.

Therefore, setting the frame size to match the estimated backlog perfectly balances and minimizes both the collision waste and the empty-slot waste.